Data-provenance trace for Research_results figures and tables

Auto-generated by `python Research_results/verify_numbers.py`. Each entry names the figure/table, the underlying CSV, the computation, and the recomputed value. If anything in a published artifact diverges from this trace, the artifact is stale — rerun `build_research_results.py`.

- Source CSV: `Research_results\data\unified_test_3way.csv`
- Matched triples (n): **523**
- Rows per agent: ReAct (GPT-5) = 523, CC-Baseline = 523, SADE = 523

Figure 1 — headline metrics (5-panel)

Source: `data/unified_test_3way.csv`, n = 523 matched triples per agent.

Panel: Overall judge (1-5)

Computation: `df[df.agent==A]['llm_judge_overall_score'].mean()`

Agent	n	mean	SEM
ReAct (GPT-5)	523	3.803	0.055
CC-Baseline	523	3.931	0.052
SADE	523	4.323	0.047

Panel: RCA F1

Computation: `df[df.agent==A]['rca_f1'].clip(lower=0).mean()`. `-1` is the runner's no-submission marker (every `loc_*` and `rca_*` metric is `-1` in the same row); we count those rows as F1 = 0 because failing to submit is a real outcome, not a missing measurement.

Agent	n_total	n_no_sub (counted as 0)	mean
ReAct (GPT-5)	523	46	0.398
CC-Baseline	523	80	0.552
SADE	523	22	0.767

Panel: Localization F1

Computation: `df[df.agent==A]['localization_f1'].clip(lower=0).mean()`. Same `-1`-as-no-submission convention as RCA F1.

Agent	n_total	n_no_sub (counted as 0)	mean
ReAct (GPT-5)	523	46	0.556

Agent	n_total	n_no_sub (counted as 0)	mean
CC-Baseline	523	80	0.680
SADE	523	22	0.776

Panel: Detection accuracy

Computation: `df[df.agent==A]['detection_score'].mean()`

Agent	n	mean
ReAct (GPT-5)	523	0.685
CC-Baseline	523	0.667
SADE	523	0.853

Figure 2 — per-fault-family overall score (heatmap)

Each cell is the mean `llm_judge_overall_score` for one (fault family × agent).

Source: `data/unified_test_3way.csv` (n = 523).

Computation per cell: `df[(df.agent==A) & (df.root_cause_name==fam)]['llm_judge_overall_score'].mean()`

Row count summary:

Fault family	n (per agent)
`arp_acl_block`	23
`arp_cache_poisoning`	23
`bgp_acl_block`	7
`bgp_asn_misconfig`	7
`bgp_blackhole_route_leak`	7
`bgp_hijacking`	3
`bgp_missing_route_advertisement`	7
`bmv2_switch_down`	4
`dns_lookup_latency`	3
`dns_port_blocked`	3
`dns_record_error`	3
`dns_service_down`	3
`flow_rule_loop`	6

Fault family	n (per agent)
flow_rule_shadowing	6
frr_service_down	13
host_crash	23
host_incorrect_dns	3
host_incorrect_gateway	13
host_incorrect_ip	23
host_incorrect_netmask	13
host_ip_conflict	20
host_missing_ip	23
host_static_blackhole	7
host_vpn_membership_missing	3
http_acl_block	9
icmp_acl_block	21
incast_traffic_network_limitation	9
link_bandwidth_throttling	21
link_detach	23
link_down	23
link_flap	23
link_fragmentation_disabled	23
link_high_packet_corruption	23
mac_address_conflict	23
mpls_label_limit_exceeded	1
ospf_acl_block	3
ospf_area_misconfiguration	3
ospf_neighbor_missing	3
p4_compilation_error_parser_state	4
p4_header_definition_error	4
p4_table_entry_misconfig	4
p4_table_entry_missing	4
receiver_resource_contention	8

Fault family	n (per agent)
`sdn_controller_crash`	6
`sender_application_delay`	9
`sender_resource_contention`	9
`southbound_port_block`	6
`southbound_port_mismatch`	6
`web_dos_attack`	9

Full numeric matrix is in `tables/table_per_family.csv`.

Figure 3 — cost-vs-correctness scatter

X-axis: `in_tokens / 1000`. Y-axis: `llm_judge_overall_score` (jittered ±0.14 for visibility, deterministic seed 42).

Each scatter dot is one row of `unified_test_3way.csv`. Total points = 3 agents × n = 1569.

Figure 4 — input-token budget boxplot

Source: `data/unified_test_3way.csv`, column `in_tokens`.

Box: median, Q1, Q3. Whiskers: 1.5×IQR. Outliers hidden.

Agent	median	Q1	Q3	mean
ReAct (GPT-5)	54,983	31,102	88,632	66,718
CC-Baseline	189,144	100,530	334,046	235,914
SADE	343,969	246,048	482,927	394,846

Figure 5 — no-submission rate

Marker: every localization/RCA metric is −1 in the row.

Computation:

```
loc_rca_cols = ['localization_accuracy','localization_precision',
                'localization_recall','localization_f1',
                'rca_accuracy','rca_precision',
                'rca_recall','rca_f1']
no_sub = (df[loc_rca_cols] == -1).all(axis=1)
```

Agent	no-submit count	total	rate
ReAct (GPT-5)	46	523	**8.8%**
CC-Baseline	80	523	**15.3%**
SADE	22	523	**4.2%**

Figure 6 — scaling by topology size

Per (agent × topology size), mean of `llm_judge_overall_score` and `in_tokens / 1000`. Topology size '-' is excluded from the figure.

topo_size	Agent	n	overall mean	in_tokens mean (k)
s	ReAct (GPT-5)	141	3.901	50.2
s	CC-Baseline	141	4.085	179.3
s	SADE	141	4.596	305.4
m	ReAct (GPT-5)	142	3.768	66.1
m	CC-Baseline	142	3.979	226.2
m	SADE	142	4.141	403.9
l	ReAct (GPT-5)	142	3.768	104.8
l	CC-Baseline	142	3.739	309.4
l	SADE	142	4.289	490.8

Figure 7 — NIKA injector limits

Two NIKA-side datasets, both run with the SADE agent fixed:

- `manual_injection`: `D:/meet/repro/nika-claude/results_manual_injection/0_summary/evaluation_summary.csv`
- `train_obs`: `D:/meet/repro/nika-claude/results_train_obs/0_summary/evaluation_summary.csv`

Panel (a) — outcome-bucket split:

Source	n	correct (≥4)	partial (2-3)	wrong (≤1)
manual_injection	39	39 (100%)	0 (0%)	0 (0%)
train_obs	39	19 (49%)	3 (8%)	17 (44%)

Panel (b) — train_obs final-outcome score histogram:

score	count
1	17
2	3
5	19

Figure 8 — time_taken per agent

Source column: `time_taken` from `data/unified_test_3way.csv`.

Agent	mean (s)	median (s)	std (s)
ReAct (GPT-5)	227.6	212.8	100.7
CC-Baseline	216.5	169.1	154.6
SADE	261.2	179.5	205.4

Figure 9 — tool-error rate (hybrid source)

Each agent uses its accurate channel:

- **ReAct (GPT-5)** — CSV `tool_errors` from `data/unified_test_3way.csv`. ReAct's LangGraph logger does not emit `is_error: true` events in the conversation log, but the runner captures errors through LangGraph's tool-error handler and records them in this column.
- **CC-Baseline / SADE** — log-counted `is_error: true` events from each session's `conversation_diagnosis_agent.log`, persisted to `data/tool_errors_from_logs.csv` by `scan_log_errors.py`. The runner's `tool_errors` CSV column reports 0 for both Claude-Code agents — the Claude Code SDK's `is_error` flag in tool_end events is not currently wired into the CSV-writing path. Bug worth fixing in the runner; for the paper we use the log truth.

Cross-check (sample, see `_verify_tool_errors.py`):

Agent	sample n	CSV `tool_errors` total	log `is_error: true` total	source we use
ReAct (GPT-5)	15	4	0	**CSV**
CC-Baseline	15	0	17	**logs**
SADE	15	0	3	**logs**

Panel (a) — sessions with ≥1 tool error:

Agent	sessions with ≥1 error	total sessions	rate	source
ReAct (GPT-5)	51	523	9.8%	CSV `tool_errors`
CC-Baseline	152	523	29.1%	log `is_error: true`
SADE	96	523	18.4%	log `is_error: true`

Panel (b) — per-tool-call error rate:

Agent	total errors	total tool_calls	error rate	source
ReAct (GPT-5)	106	13,266	0.80%	CSV `tool_errors`
CC-Baseline	454	13,962	3.25%	log `is_error: true`
SADE	158	10,393	1.52%	log `is_error: true`

Figure 10 — tool-call efficiency

Source column: `tool_calls` from `data/unified_test_3way.csv`. Verified against `tool_start` event counts in the per-session `conversation_diagnosis_agent.log`: 30 random sessions sampled, all 30 matched exactly (see `Research_results/_verify_tool_calls.py`).

Panel (a) — Mean tool_calls per session:

Agent	mean	SEM	std
ReAct (GPT-5)	25.37	0.51	11.66
CC-Baseline	26.70	0.62	14.16
SADE	19.87	0.39	8.89

Panel (b) — tool_calls per correct submission:

Computation: `df[df.agent==A]['tool_calls'].sum() / (df[df.agent==A]['llm_judge_final_outcome_score'] >= 4).sum()`

Agent	total tool_calls	correct (final≥4)	calls per correct
ReAct (GPT-5)	13,266	268	49.5
CC-Baseline	13,962	271	51.5
SADE	10,393	382	27.2

Table 1 — Headline metrics (`tables/table_headline_metrics.md`)

Cell formulas:

```
# Correctness columns (Overall, Final, Detection):
df[df.agent==A][col].dropna().mean()

# F1 columns (Localization F1, RCA F1) — no-submission rows count as 0:
df[df.agent==A][col].clip(lower=0).dropna().mean()

# No-submission rate — fig05 marker (all 8 loc/rca metrics == -1):
loc_rca = ['localization_accuracy','localization_precision','localization_recall',
```

```
              'localization_f1','rca_accuracy','rca_precision','rca_recall','rca_f1']
(df[df.agent==A][loc_rca] == -1).all(axis=1).mean()
```

Already covered by Figure 1 / 4 / 5 / 8 / 10 entries above; numeric values match.

Table — time_efficiency (`tables/table_time_efficiency.md`)

Agent	mean time	median time	mean tokens	% correct (final≥4)	tokens / correct
ReAct (GPT-5)	227.6	212.8	66,718	51.2%	130,199
CC-Baseline	216.5	169.1	235,914	51.8%	455,287
SADE	261.2	179.5	394,846	73.0%	540,588

Table — topology_scaling (`tables/table_topology_scaling.md`)

Mirrors fig06; values in the figure trace are correct.

Cross-cutting integrity checks

Check	Expected	Actual	OK
Σ rows per agent == total rows	1569	1569	✓
All three agents have equal row count	all == 523	[np.int64(523), np.int64(523), np.int64(523)]	✓
time_taken has no NaN	0	0	✓